

Ancient water system and reservoir at the Muarajambi archaeological site

C.M. Arkhi & W.R. Wahyudi

Department of Archaeology, Faculty of Humanities, Universitas Indonesia, Depok, Indonesia

ABSTRACT: *Muarajambi* is an archaeological site that had experienced numerous events of archaeological ruins in the period of 9th–12th century AD. The structural ruins at this site showed the influence of Buddhism on both the layout and art in the structures, buildings and sculptures. On the basis of the influence of Buddhism on these premises, this site was considered a Buddhist archaeological site. This claim was supported by the latest theory, according to which the Muarajambi archaeological site was a Mahavihara, that is, a Buddhist teaching centre in Nusantara. The theory was developed on the basis of the archaeological ruins and the historical information collected from the journals written by I-Tsing in the 7th Century AD. I-Tsing was a Buddhist monk who travelled from China to India and lived in Muarajambi for 6 months. Thus, we can assume that the greater area of the Muarajambi site was also populated by quite a number of people and, therefore, it required some type of water resource to support the activities taking place inside the Mahavihara. The hypothesis that the site was located near a water resource could be true because of the presence of a river watershed near the site. In addition, there were traces of water channels, ponds and reservoirs on the site, all of which were likely built by the monks who lived there in the past.

1 INTRODUCTION

As an archaeological site, Muarajambi has a great and varied collection of archaeological findings that have been originated from the classical archaeology era of Indonesia. Some of the artefacts included beads, potteries, ceramics, bronze structures, stone statues, slabs, brick buildings or structures and earth-mounds, which may have hid the ruins of brick structures (commonly referred to as *menapo* in the local language). The most recent research conducted on the subject was presented in a dissertation by Agus Widiatmoko (2015) entitled “*Muarajambi Site as the Mahavihara from VII—XII AD*”, which compares the ruined archaeological sites in Muarajambi with the archaeological sites in Nalanda and Wikramasila in India. Widiatmoko used a historical archaeology method and a historical analogy technique simultaneously by analysing a documented records of a Chinese monk named I-Tsing, written during his pilgrimage from China to India to study the true teachings of Buddhism in its birthplace. The author also used a similar approach for this study.

The information collected from the documents of I-Tsing mentioned a stopover in a town on the island of *Suvarnadvipa* (historians and archaeologist agreed that *Suvarnadvipa* is the Sanskrit name for the island of Sumatera in the past) before heading to India. The town was called Fo-Shi. I-Tsing mentioned that he sailed down a river to reach Fo-Shi, which may indicate that the town was located in the hinterland part of the island and there might be a watershed nearby. He also mentioned that the town was surrounded by a fortress and there were thousands of Buddhist monks practicing the teachings of Buddha as in Madhyadesa, India. It is evident from the documents of I-Tsing that thousands of Buddhist monks studied in Fo-Shi; therefore, it could be assumed that the town had facilities to support their lives and activities.

To date, several archaeological sites with Buddhist characteristics have been found in Sumatera, such as the Padanglawas site in North Sumatra Province, the Muaratakus site in Riau

Province and the Muarajambi site in Jambi Province. I-Tsing's further narrowed down to two archaeological sites from the classical era with a high concentration of archaeological ruins located near watersheds, the Batanghari River and the Musi River. However, more number of archaeological ruins were found in the site located near the Batanghari watershed, specifically in the Muarajambi area. Archaeological findings in this site showed strong Buddhist characteristics either in written objects, such as the Buddhist mantras inscribed on the golden plates, or in the form of a Buddhist figurine of *Prajnaparamitha*, and curved structures of stupas made of bricks that can be found in several archaeological structures. This fact assured earlier researchers and writers that the ruins in the Muarajambi site are the archaeological remains of Fo-Shi from the past.

1.1 *Problem formulation*

Muarajambi is an attractive landscape, surrounded by ponds, lakes and streams. In his documents, I-Tsing mentioned a tradition followed by the monks in Fo-Shi that resembles a tradition practiced in India. If this was true, then it was possible that the daily activities performed by the monk community in Fo-Shi were similar to the ones practiced in Nalanda monastery, India. Furthermore, he cited the routines performed by the monks in Nalanda monastery and the tradition of building bath ponds, which was used by the monks every morning upon waking up (Takakusu, 2014: 241–242). I-Tsing also mentioned other activities such as washing hands after a meal and screening insects from the water obtained from the water source. These assumptions were confirmed by the fact that many of the archaeological findings were located near rivers and other water sources.

In archaeology, different approaches are followed for handling or reviewing a particular landscape. Historical archaeologists are often divided between the natural environment and artificial environment. This fact is probably based on the positivist perspective that always considers the interactions between humans and the natural environment as a process of environmental adaptation. Later, from the viewpoint of hermeneutics, we also have a differentiated the concepts of 'space' and 'place'. 'Space' can be interpreted as the environment where people live, whereas 'place' is the space that has been interpreted by humans (Hoed, 2011, p. 110).

The proximity to the archaeological ruins and the landscape of Muarajambi provide a base for the assumption of associations between water and the presence of streams, ponds and lakes. From the viewpoint of hermeneutics, we aim to conceive the landscape as a 'place' that has significance through the memory and values constructed by the past inhabitants. To achieve this objective, the author proposed the formulation of the research problem in the following statements:

1. What is the significance of the proximity between the archaeological remains and the water channels and reservoirs in Muarajambi?
2. What is the significance of water with respect to religious activities of the Fo-Shi monks in the past in Muarajambi?

2 RESEARCH METHODS

In general, archaeology is the study of life in the past using documented archaeological evidence (Dark, 1995, p. 1). However, not all material evidence (artefacts) found in an archaeological research can be regarded as archaeological data. Only the evidence of archaeological material that has information related to the one found later can be regarded as archaeological data. Furthermore, archaeological data must first clear an interpretation framework to be considered as archaeological evidence (Dark, 1995, p. 36), that is, archaeological data in a study should be placed within a systematic framework so as to justify it scientifically. In this case, the new archaeological data can be interpreted only if it is placed in the right context, in a good spatial context, time context, as well as cultural context. By following this sequence, the research can be divided into four phases, namely:

1. Determination of data sources
2. Collection of archaeological information on the data source
3. Establishment of the archaeological evidence
4. Interpretation.

The first stage of determining the source of data was carried out by reviewing the available literature works on the subject, the findings from previous studies as well as the restrictions on the problems, the formulation of the coverage area of the research and the availability of data in the area of research. This was done to segregate data resources that can only be obtained as a set or in a singular form (datum), which would later be used in this archaeological research. This study included data on water channels, ponds, reservoirs and the ruins—found in the archaeological area of Muarajambi.

After determining the data sources, the relevant archaeological information was extracted from them through describing the location and the shape and size of the data unit. This description was made to obtain an important aspect of the research, namely shaping the context of the archaeological data from its provenience. As mentioned by Dark, context is important in understanding archaeological data. To obtain a context, an archaeologist should determine the significant relationship between the findings and the surrounding environment (Hodder, 1991, p. 43). Therefore, at this stage of the formation of archaeological evidence, we processed the archaeological information obtained from the data source and combined them with other related archaeological data. This was done by plotting the archaeological features found along the waterways. The subjects in this case are the ruins of ancient brick structures and the *menapo*. A similar procedure was carried out on the artefacts located along or near the ponds and reservoirs in the Muarajambi site. Therefore, by effective engineering, we could distinguish potential water sources from less potential water sources. After obtaining the archaeological data, the next phase is interpreting them as archaeological evidence.

One of the approaches used in the data interpretation process for this type of research is analogy, which is generally used to describe an unknown object by comparing it to another object with similar identifiable characteristics for reference (Sharer & Ashmore, 2005, p. 455). In other words, an analogy refers to a comparison of two identical objects, between the archaeological data and reference objects, by focusing on their similarities. The analogy process used in this study is direct historical analogy, the purpose of which is to describe the identity or the relationship between the archaeological data and the reference object used in the comparison by observing the common traits shared by the object. Furthermore, the particulars of the activity are recorded in the historical sources (Gamble, 2001, p. 87). In this study, the author compared the archaeological evidence and the description provided by I-Tsing in his documents.

2.1 *Environment description*

Administratively, the Muarajambi archaeological site is located in the district of Muarajambi, Jambi Province, at approximately 40 km from the city of Jambi. In 2013, this area was designated as the National Heritage Area by the Regulation of the Minister of Education and Culture (Kepmendikbud No.259/M/2013). Geographically, the region of Muarajambi is located on the banks of the Batanghari River with an altitude of 8–12 m above sea level. Astronomically, the region of Muarajambi lies at 1° 24' South Latitude–1° 33' South Latitude and 103.22' East Longitude–103.45" East Longitude (Geographic Coordinate System) or 48 M 346839 mT–356779 mT; 9833468 mU–9840765 mU (Universal Transverse Mercator). The site is located in the area of a natural dike (flood plain) where floods flow from alluvial deposits from the Batanghari River (BPCB Jambi, 2004, p. 7). The physiography of Muarajambi mostly consists of a gently sloping plateau (67.72%), followed by choppy and bumpy hills. Most areas of the plains in Muarajambi are at an altitude of 10–100 m above sea level (74.95%), and only a few areas (4.50%) have an altitude >100 m above sea level (BPCB Jambi, 2004: 9). The level of surface water is higher in the eastern part than the western part because of the topographic nature of the land; the eastern part of Muarajambi is a basin and a swampy area and therefore water does not get absorbed immediately by the soil and it flows

as a runoff. The stream patterns in Muarajambi, depending on the intensity of the rainfall, is secondary to the risk posed by stream floods that occur when the main river cannot support the water tributaries (BPCB Jambi, 2004, p. 10). Therefore, we may conclude that, in general, Muarajambi is a lowland area that is prone to floods.

2.2 Data

Data obtained in this study can be grouped into the following three categories: water channels, ponds and reservoirs. The following is the description of the terms used in this article.

2.2.1 Water channels

In this study, a water channel is defined as a long basin of water flowing through the archaeological area of Muarajambi. The 17 water channels and their local names (see the list and map in the appendix) in the region are grouped into the following classifications of *parit*, *buluran* and *sungai*. The distinction in the local names is most likely associated with the width of the stream; however, obtaining the true meaning of their names is beyond the scope of this study and may require more studies involving toponymy or other similar approaches.

2.2.2 Ponds

In general, a pond refers to a broad alcove on the ground, which contains water. In this article, a pond is considered a niche or a recess on the ground, where a small amount of water is collected. In the area under study, there are 10 ponds (see the list and map in the appendix).

2.2.3 Reservoirs

A reservoir refers to a huge water storage used for satisfying human needs or managing water distribution. In this study, reservoirs refer to large water bodies located in the archaeological area, including *dano* (lake), *payo* and *rawa* (local terms). In the study area, there are six reservoirs (see the list and map in the appendix).

2.2.4 Features

As previously mentioned, Muarajambi is an archaeological site where several evidences of archaeological ruins have been found, including some large structures. In general, the archaeological structures with ruins found in the area were made of bricks. To date, approximately 12 archaeological structures and 69 *menapo* have been found and registered by the BPCB (*Balai Pelestarian Cagar Budaya/the Agency for the Preservation of Cultural Heritage*) Jambi (see the list and map in the appendix).

2.3 Data association

The framework of interpreting the historical analogy was obtained by understanding the value of and information found in the archaeological evidence. For this purpose, the author used relevant historical sources, including the travelogues of I-Tsing, which contained some information related to the tradition of Buddhism practiced in his motherland, China, countries of the southern island, Fo-Shi, as well as India. Among the many traditions observed by him, I-Tsing claimed that the traditions practiced by the monks in Fo-Shi matched those practiced in India. He noted that the monks in Fo-Shi maintained conformity with the sutras (*a type of religious literature*) similar to that of India, which is evident from his documents. Regarding water, I-Tsing mentioned that there were two types of water for daily activities, namely water for drinking (*Kundi*) and water for rinsing (*Kulasa*) (Takakusu, 2014, p. 130). Both could be obtained from wells, ponds or streams, according to the rules written in the Book of Discipline or the *Vinaya Sutra*. The synchronic phenomenon was demonstrated by combining the archaeological data and the description shown by the historical data taken from the documents of I-Tsing. Therefore, they can be elaborated in the following discussion.

2.4 Association of water channels with the features

The first step in limiting the geographical scope of processed data in this study was grouping the water channels with potentially stronger linkages with the archaeological features. This was done by re-grouping the remaining water channels located at a distance of ± 100 m from the banks of the waterways. This argument is based on the assumption that human activities related to water could not be carried out too far from the water source. The total number of features after re-grouping is listed in Table 1.

The highest number of features for a water channel indicates that the particular channel has potential association with human activities related to water in the past. The channels with the highest number of features are Sungai Jambi, Sungai Parit Sekapung and Parit Johor. Thus, the next set of water channels which also have potential association with human activities directly intersect with the aforementioned three channels. They also have a number of features distributed along the vicinity (see the map in the appendix), including Sungai Terusan, Sungai Melayu, Buluran Paku and Buluran Dalam. There are also other channels with connecting drains to the trenches that surround a feature, such as Parit Candi Sialang (which is connected to the trench around Candi Sialang) and the Parit Koto Mahligai (connecting Sungai Terusan Channels to the trench of Candi Koto Mahligai). Moreover, the water flow in some channels shows an anomaly of a reverse flow at some points, such as the inflows in

Table 1. Total number of features.

No.	Water channel	Total number of features in the range of ± 100 m
1	Buluran Paku	7
2	Parit Bukit Perak	1
3	Parit Candi Sialang	1
4	Parit Sekapung	14
5	Sungai Amburanjalo	2
6	Sungai Berembang	0
7	Sungai Danau Ulat	0
8	Sungai Kemingking	5
9	Sungai Medak	0
10	Sungai Selat	2
11	Sungai Selat II	0
12	Sungai Seno	0
13	Sungai Terusan	1
14	Parit Johor	13
15	Sungai Jambi	34
16	Sungai Melayu	2
17	Buluran Dalam	1

Further details can be found in the appendix.

Table 2. Interconnected water channel(s).

No.	Reservoir in the Muarajambi area	Interconnected water channel(s)	Total number of interconnected water channel(s)
1	Rawa Buluran Kli	Parit Johor	1
2	Dano Kelari	Sungai Jambi and Sungai Selat	2
3	Dano Ulat	Sungai Selat II	1
4	Payo Rimbo Terbakar	Sungai Melayu	2
5	Payo Rawang Bato	Parit Sekapung	1
6	Payo Terjun Gajah	—	0

Further details can be found in the appendix.

Sungai Selat from the Batanghari River. The reverse flow also occurred in the intersections of Sungai Terusan, from Parit Sekapung to Sungai Jambi channel, and it might indicate human intervention or human engineering in the past.

Another category of the water channel in the Muarajambi area which the author suspects to have an association with human activities is the reservoirs. In general, the region of Muarajambi is a swampy area, which is prone to floods; therefore, if the Batanghari River overflowed during heavy rainfall, then the water will simply go to the watersheds. Every reservoir in the area has a connection to one or two water channels, as illustrated in Table 2.

Because of this phenomenon, reservoirs found in the Muarajambi area might be naturally formed or artificially constructed to address similar issues. Nevertheless, compared to the results of earlier analysis on the waterways combined with data of the features, hypothetically, there should be a network of dams in the Muarajambi area intersecting with the water channels, forming a potential association with community activities in the past.

2.5 Association of features with ponds

The presence of ponds around Muarajambi is also of interest for archaeological findings. Some of the features were found within a radius of ± 100 m around the ponds. The use of the ± 100 m range as the perimeter in the analysis of features–ponds association is related to the earlier analysis. The results are presented in Table 3.

The above-presented recapitulation results show that some ponds have potential associations with the features due to their proximity, namely Kolam Parit Johor I, Kolam Parit Johor II, Kolam Telago Rajo, Kolam Sangkar Ikan, Kolam Sungai Jambi, Kolam Candi Tinggi and Kolam Pemandian Ayam 2. Another argument to support the author's hypothesis of the presence of humans around these ponds is the aspects related to their size, length and width, which seemed to support the fact that any activity requiring water would be carried out as close as possible to the abundant water sources. This conviction was also supported by the findings from an earlier hydrological study conducted by Sutikno, Aris Poniman and Maulana Ibrahim revealing that the base of Telago Rajo ponds (Kolam Telago Rajo) were cut crossing the layer of groundwater surface, so the overflow of groundwater would be directed to the pond. This finding provides strong evidence that Kolam Telago Rajo is manmade (Sutikno, 1992: 117). After reviewing the study cited above, we might surmise that a pond or *kolam* with a similar shape and size to Kolam Telago Rajo might also be artificially made in the past.

The documents of I-Tsing about the routines of the monks in Nalanda monastery, India, mentioned a tradition of building a bath pond of the size of 1 *yojana* (15 km); there were about 20–30 ponds with the size of 1 *mu* (0.84 m²) to 5 *mu* (4.2 m²) (Takakusu, 2014: 241–242). The monks would go to these ponds every morning with a bath cloth (because nudity is prohibited in Buddhism). On the basis of an analogy about traditions performed in Fo-Shi and India with

Table 3. Association of features with ponds.

No.	Ponds in muarajambi	Amount of features in the range of ± 100 m
1	Kolam Parit Johor I	4
2	Kolam Sungai Selat	0
3	Kolam Sungai Jambi	2
4	Kolam Candi Tinggi	2
5	Kolam Parit Johor II	4
6	Kolam Pemandian Ayam	0
7	Kolam Pemandian Ayam 2	1
8	Kolam Sangkar Ikan	2
9	Kolam Telagorajo	4
10	Kolam Tanjaan Galah	0

Further details can be found in the appendix.

regard to the liturgy, we may surmise that the reason for adjacent arrangement of ponds as seen in the features in Muarajambi was to follow the traditions of pond-making in India.

3 CONCLUSION

Historical data in the form of travelogues from I-Tsing, a Chinese scholar, have provided the information and reference in the research on the archaeological ruins at the Muarajambi site. By using the hermeneutics approach that had previously been referred to in the introduction of the archaeological environment, the archaeological site is considered to have not only a 'space' but also a 'place', which support the values of the monks in the past. Therefore, on the basis of this fact, we may draw the following conclusions.

The waterways, ponds and reservoirs in Muarajambi were of significance for the monks who lived at the time, because they provided the means for performing their religious activities and routines.

The water channels that were closely associated with the activities of the monks in the past were Sungai Melayu, Buluran Paku, Buluran In, Parit Temple beehive and the branching river channels to the temple features of Koto Mahligai.

Reservoirs intersecting the water network of channels in the second point are closely associated with the function of the water channels. In addition, reservoirs in the region support the management system in overcoming the outburst of floods during heavy rainfall in Muarajambi.

Clearly, Kolam Telago Rajo is an artificially constructed pond, the shape and size as well as the historical data of which imply the presence of other similar artificial ponds in the area.

The Muarajambi site, which has been designated as a National Heritage Area, is one of the major references in the history of Buddhism in the archipelago, as demonstrated by the numerous archaeological ruins in the area. The author believes that this study will lay the foundation for further studies in this area, and using scientific tools, we will learn more about the reversal of water flow and the integrated water networks in the Muarajambi site.

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